Uhle Box Dewatering & Sizing

Due to so many system variables, each felt dewatering system requires special attention as
"What works for one machine may not work for another"

If the Uhle box is under or over sized it will make the vacuum system inefficient. With an inefficient felt dewatering system, a felt could run too wet after the Uhle box and into the press nip causing sheet problems such as breaks and crushing.
It is much easier to incorporate the right size Uhle pipe into a paper/board machine than to fix it into an existing press section. When making a change to an existing felt dewatering system the vacuum source capacity must be known or selected. Without information concerning the vacuum source or reviewing present machine Uhle box performance, a change in the existing **Uhle box slot size** and **top cover design** could cause a major system problem.
Another important consideration in designing a felt dewatering system is the felt construction, basis weight and permeability. As the felt basis weight increases, the moisture handled by the felt also increases. An increase in felt permeability (a more open area) will allow more air to pass through the felt at a given pressure drop (vacuum level). The same is true for the opposite conditions, ***less weight carries less water and lower permeability allows less air to pass through the felt.***
Both felt basis weight and felt permeability are key inputs to the dewatering equations published in **TAPPI – TIS – 0404 – 27**. These equations can be used to calculate Uhle box cover open areas. These dewatering equations take into account the water content in the felt as well as the machine speed. Water content is given as **kg of water/kg of felt** and **dwell time** is given in milliseconds. Dwell time is defined as the time which the felt is over the Uhle pipe cover open area. For good Uhle pipe dewatering system the **dwell time should be between 3 and 6 milliseconds**, however for **machine speeds less than 229 m/min**, a **0.5"** (12.7 mm) slot width is the minimum which should be used. Felt dewatering is truly a function of airflow air velocity.
Keeping these factors in mind it is easy to understand why adding an additional Uhle pipe, changing a cover style, changing cover open area, or changing a vacuum source can have a significant effect on machine runnability.

INDIVIDUAL SLOT WIDTH LESS THAN ½" (12.7 mm) HAS A TENDENCY TO PLUG
&
INDIVIDUAL SLOT WIDTH WIDER THAN 1" (25.4 mm) WILL INCREASE FELT WEAR

* **Factors effecting suction pipe sizing:-**

1. Air flow and vacuum applied.
2. Felt type.
3. Felt construction.
4. Water content of felt.
5. Dwell time (slot design).

Calculations for required slot width: -

If,	face length	= 2.85 meter = 2850 mm
	machine speed	= 420 m/min
	required dwell time	= **3 milliseconds** for successful water removal
then,	slot width (mm)	= [m/c speed (m/min) x dwell time (milliseconds)] / 60
		= [420 x 3] / 60
		= 1260 / 60
		= **21 mm**, which is not greater than 1" and to use in **single suction tube**?

therefore, the Uhle box slot size = 2850 x 21 mm

similarly,

If, face length = 2.85 meter = 2850 mm
machine speed = 420 m/min
required dwell time = **4 milliseconds** for successful water removal
then, slot width (mm) = [m/c speed (m/min) x dwell time (milliseconds)] / 60
= [420 x 4] / 60
= 1680 / 60
= **28 mm**, which is greater than 1" and to use in **two suction tubes**?

Therefore, the two individual Uhle boxes slot sizes = [2850 x 14 mm] & [2850 x 14 mm] or [2850 x 13 mm] & [2850 x 15 mm]

Typical inlet moisture values are:-

1st Press Felt = **0.1** kg **water** / kg **felt**
2nd Press Felt = **0.8** kg **water** / kg **felt**
3rd Press Felt = **0.6** kg **water** / kg **felt**

for this values the felt shower water pressures to be controlled.

Calculation for required pump capacity:-

Vacuum Pump Capacity (m^3/min) = Slot Area (m^2) x Vacuum Factor (m^3/min/m^2)
Vacuum factor ranges from **660** to **970 m^3/min/m^2**
If , we take the average factor of 700 m^3/min/m^2
Then , vacuum capacity = [0.021 x 2.85 m^2] x [700 m^3/min/m^2]
= 41.895 m^3/min for one press felt Uhle box
hence , for 3 press felt Uhle boxes the required **vacuum pump capacity** = 41.895 x 3

= **125.685 m^3/min**

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